Week 13 Seminar: Understanding Confounding and Causal Inference

Conditioning vs. Design-Based Approaches

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Objectives

- Differentiate between **conditioning-based** and **design-based** approaches to solving confounding.
- Identify confounding variables that each approach addresses or leaves unaddressed.
- Discuss the assumptions needed to infer causality.
- Apply insights to the Nike Vaporfly running shoe example.

Background Reading

Before the Seminar: Students should have read:

- *"tutorial_researchDesign_reading"* on Canvas.

Today's Task:

- Group discussions to classify study designs and confounders.
- Draw conclusions on strengths, limitations, and assumptions of different methods.

Example Context: Nike Vaporfly

Research Question: Do Nike Vaporfly shoes improve running performance? **Challenges:**

- Self-selection: Runners choose whether to wear Vaporfly.
- External factors: Race conditions, training, and sponsorship bias.
- Lack of randomized controlled trials.

Conditioning vs. Design-Based Approaches

Conditioning-Based:

- Use statistical models to adjust for observed confounders.
- Example variables: Age, gender, training history, weather.

Design-Based:

- Leverage natural experiments or repeated observations to control for unobserved confounders.
- Examples: Difference-in-Differences (DiD), within-runner comparisons.

Difference-in-Differences Logic

- Compare performance changes over time:
  - **Treatment Group**: Runners switching to Vaporfly.
  - **Control Group**: Runners not switching shoes.
- Controls for:
  - *Time-invariant factors:* Runner's innate ability.
  - *Common trends:* Weather, course difficulty.
- Causal Effect = Difference in changes between groups.

Mathematical Representation:

- Let Y_{it} be the running performance of individual i at time t.
- The Difference-in-Differences estimator can be represented as:

$$\Delta Y_{\text{treatment}} - \Delta Y_{\text{control}} = (Y_{\text{post, treatment}} - Y_{\text{pre, treatment}})$$

- $(\mathrm{Y}_{\text{post, control}} - Y_{\text{pre, control}})$

- Assumes that in the absence of treatment, the average outcome for both groups would have followed parallel paths.

Boston Marathon Example

Data: Boston Marathon results from 2017 and 2018.

- **2017 (Pre-Vaporfly)**: Similar performance between groups.
- **2018 (Post-Vaporfly)**:
  - Vaporfly adopters improved more than non-adopters.
  - Causal effect attributed to Vaporfly.

Key Assumptions for DiD

- **Parallel Trends:** Without Vaporfly, both groups would have similar trends over time.
- **No Unobserved Time-Varying Confounders:**
  - Example of violation: Adopters increase training intensity more than non-adopters.
- If assumptions hold, DiD isolates causal effects.

Strengths and Limitations

Strengths:

- Controls for runner-specific and common temporal factors.
- Robust to various statistical methods.

Limitations:

- Selection bias: Vaporfly adopters may be inherently more competitive.
- Non-parallel trends could bias results.

Group Exercise Instructions

Based on: Canvas reading *"tutorial_researchDesign_reading"*

Task: For Each Approach, Answer the Following:

1. **Type of Approach:** Is it *conditioning-based* or *design-based*?
2. **Confounding Variables:**
   - What is addressed?
   - What remains unaddressed?
3. **Key Assumptions:** What assumptions are necessary for causal inference?

Goal:

- Compare insights across groups.
- Discuss trade-offs, strengths, and limitations of each approach.

Design 1: Conditioning-Based Approach

Controlled Variables:

- Age, gender, race history, prior training, distribution of race times (race hardness), weather conditions.

Limitations:

- Does not address improvements in training techniques or adoption due to sponsorships (e.g., Nike shoes).
- Selection bias: Runners using the shoes may differ from those who do not.

Key Assumption:

- No systematic differences between users and non-users of the shoe after controlling for observed variables.

Design 2: Difference-in-Differences (Diff-in-Diff)

Advantages:

- Compares “treated” (with shoes) and “control” (without shoes) pre- and post-change.
- Accounts for:
  - Unchanging features of runners (e.g., innate ability).
  - Race-level factors affecting all runners equally (e.g., weather, course difficulty).

Limitations:

- Fails to control for time-varying confounders affecting adoption of new shoes (e.g., increased training intensity by adopters).

Key Assumption:

- Trends over time for shoe adopters and non-adopters are similar, apart from the effect of the shoes.

Designs 3/4: Same Case Over Time

Overview:

- Focus on the same runners across multiple races to track changes over time.
- Provides insights into individual-level changes associated with shoe adoption.

Limitations:

- Still subject to time-varying individual confounders (e.g., training effort, health changes).

Assumptions:

- Changes over time unrelated to shoe adoption are consistent across individuals.